

IMPORTANT CHANGE

WRITTEN EXAM 1

OCTOBER 20, TUESDAY 11.40

IN THE MACHINE LEARNING SLOT

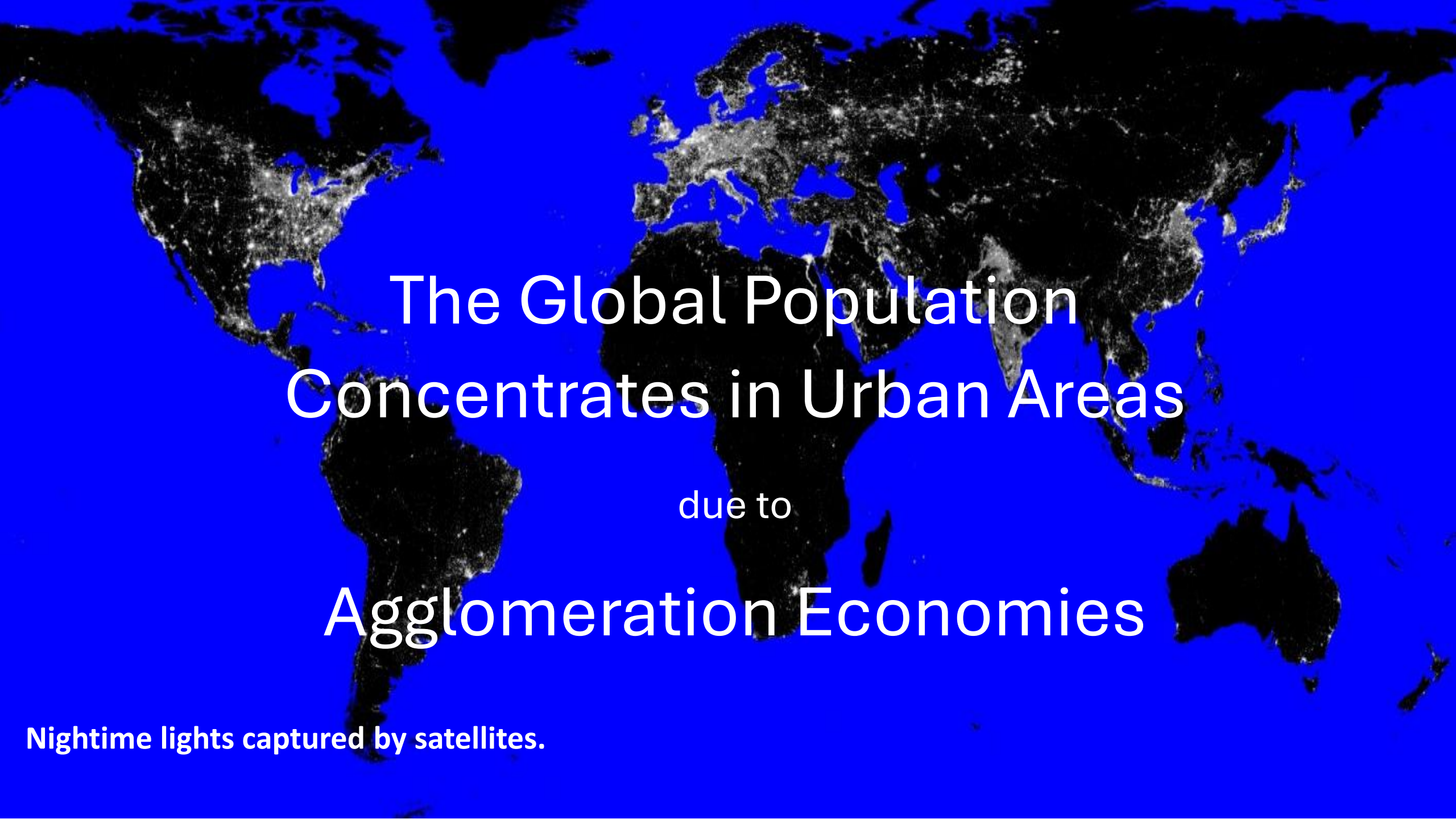
(INSTEAD OF OCT 21, WEDNESDAY)

The development of cities and regions 2: agglomeration, and the specialization-diversity debate

Balázs Lengyel

Economic Complexity, Class 3 – Sept 30 2026

Social Data Science, Corvinus University of Budapest

A world map with a black background, where landmasses are outlined in white. The map is filled with numerous small, bright white dots and clusters of dots, representing nighttime lights captured by satellites. These lights are most concentrated in North America, Europe, and East Asia, with smaller clusters in South America, Africa, and Australia. The text is overlaid on the map.

The Global Population Concentrates in Urban Areas

due to

Agglomeration Economies

Nighttime lights captured by satellites.

Today

1. Agglomeration economies
2. Concentration of knowledge and innovation
3. Specialization versus diversity

1. Agglomeration economies

Definition and sources

Agglomeration Advantages:

- Firms become **more productive** thanks to agglomeration benefits.
- Growth is **faster in areas** where **positive externalities** are present.

Sources of agglomeration advantages (*Puga, 2010*):

- **Sharing:** use of labor and urban infrastructure
- **Matching:** meeting of supply and demand (e.g., employers and employees)
- **Learning:** knowledge spillovers and learning among firms

Static and dynamic agglomeration advantages:

- **Neo-Marshallian view:** local externalities improve firms' **static performance** (efficiency, cost)
- **Neo-Schumpeterian view:** local externalities enhance **innovative capacity**

Krugman: New Economic Geography

Explanation of Industry Concentration

- Increasing returns: stimulate agglomeration
- Transportation costs: encourage proximity to raw materials and intermediate goods
- Worker migration: increases labor supply and local market size



Paul Krugman (Princeton)
Nobel prize 2008

Krugman: centrum-periphery

Initially, equal distribution between the two regions: relocation of a firm has opposing effects

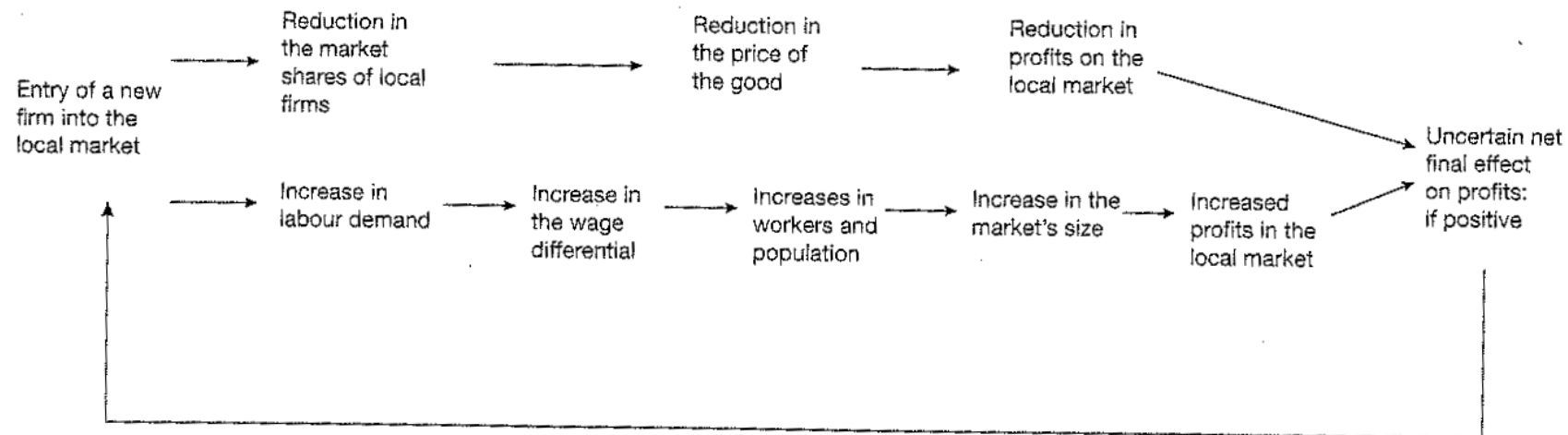
Competition effect:

- Depends on the **substitutability of products** (higher elasticity → greater profit loss)
- Also influenced by **transportation costs** (a decrease → lower prices → increased competition)

Demand effect:

- Driven by **increasing returns** (higher profit margins, stronger attraction for firms)
- Influenced by **income spent on products**

(a) The 'competition effect' and the 'demand effect'



High transportation costs:

- Stable equilibrium: firms do not relocate

Low transportation costs:

- Stable equilibrium: the industry concentrates in one region (where agglomeration advantages are greater)

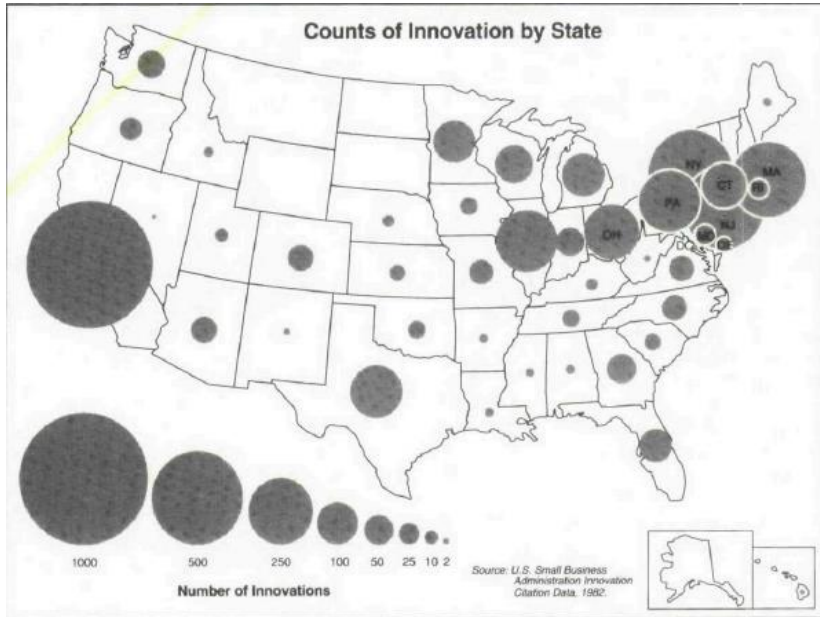
Moderately high transportation costs:

- Both equal distribution and core-periphery structure can be stable equilibria



2. Concentration of knowledge

Audretsch-Feldman 1996 – American Economic Review



- Concentration of technological innovation
- Spillovers are local
- Data: new products published in technological journals

- Gini index

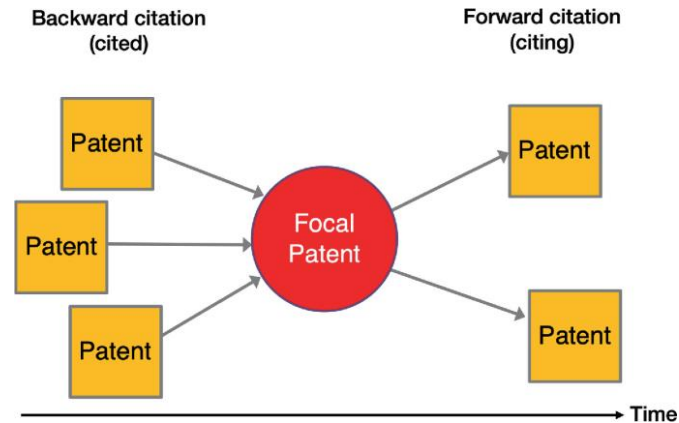
$$G = \frac{1}{2\bar{x}n^2} \sum_i \sum_j |x_i - x_j|$$

It compares the **average deviation** of an industry's share in a given region from its share in all other regions, **relative to the overall average**.

The larger the value, the more the industry is **concentrated in a few regions**.

- Comparison between production's and innovation's concentration
- Explanatory variables: natural resources, economies of scale, transportation costs, skilled workforce, R&D intensity, university R&D

Distance in knowledge spillovers



GEOGRAPHIC LOCALIZATION OF KNOWLEDGE SPILLOVERS AS EVIDENCED BY PATENT CITATIONS*

ADAM B. JAFFE
MANUEL TRAJTENBERG
REBECCA HENDERSON

We compare the geographic location of patent citations with that of the cited patents, as evidence of the extent to which knowledge spillovers are geographically localized. We find that citations to domestic patents are more likely to be domestic, and more likely to come from the same state and SMSA as the cited patents, compared with a “control frequency” reflecting the pre-existing concentration of related research activity. These effects are particularly significant at the local (SMSA) level. Localization fades over time, but only very slowly. There is no evidence that more “basic” inventions diffuse more rapidly than others.

The Quarterly Journal of Economics, Volume 108, Issue 3 (Aug., 1993), 577-598.



Journal of Urban Economics
Volume 42, Issue 3, November 1997, Pages 422-448



Regular Article

Local Geographic Spillovers between University Research and High Technology Innovations ☆

Luc Anselin ^a, Attila Varga ^{b c}, Zoltan Acs ^{d e}

Abstract

This paper re-examines the empirical evidence on the degree of spatial spillover between university research and high technology innovations. The familiar Griliches–Jaffe knowledge production function is estimated at both the state and the metropolitan statistical areas (MSA) level and extended with more precise measures of spatial spillover. Alternatives based on the gravity potential and covering indices are formulated for Jaffe's “geographical coincidence index” and found to provide strong evidence of local spillovers at the state level. At the MSA level, a distinction is made between research and development activities and university research in the MSA and in the surrounding counties. Evidence is found of local spatial externalities between university research and high technology innovative activity, both directly and indirectly via private research and development.

The gravity law

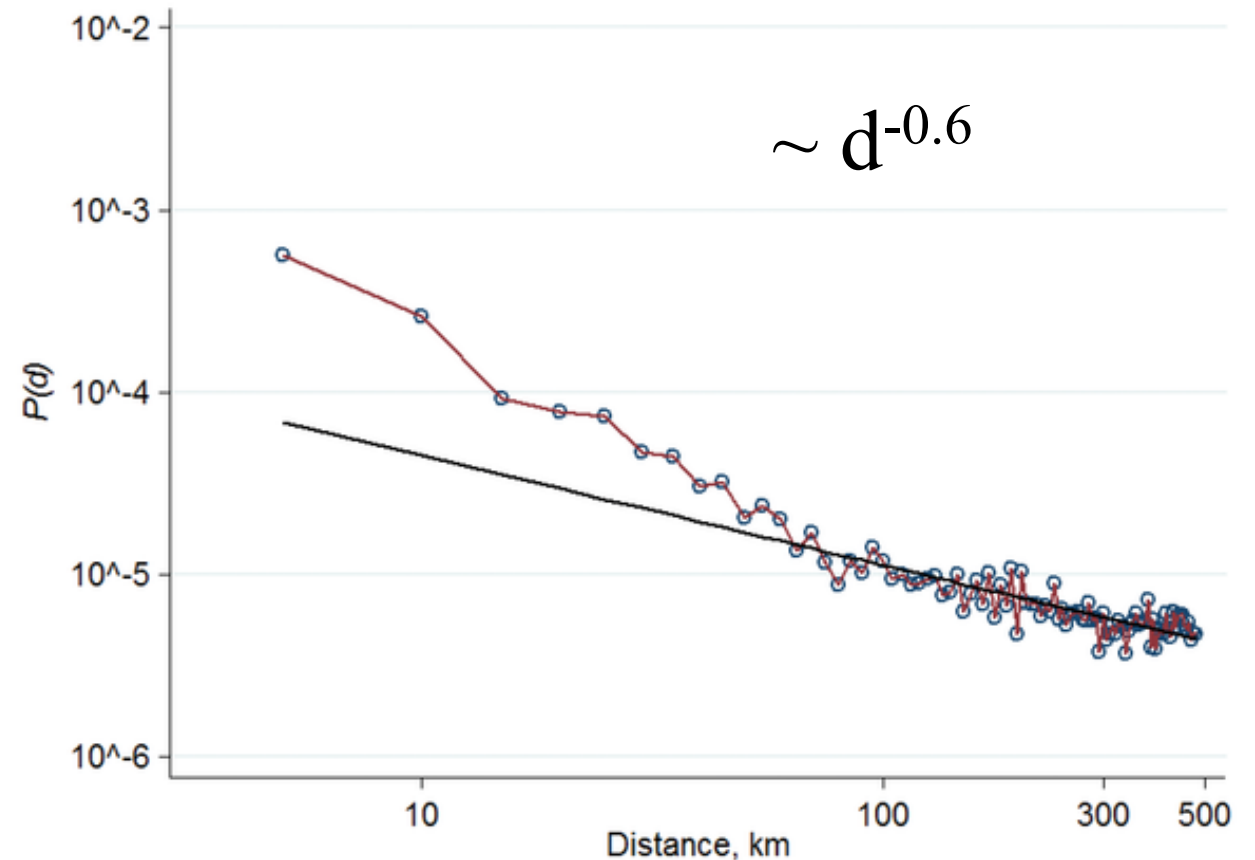
Zipf: $T_{ab} = K(P_a^\alpha P_b^\beta) / d_{ab}^\gamma$

Probability of relationships:

$N(d)$ – all possible connections

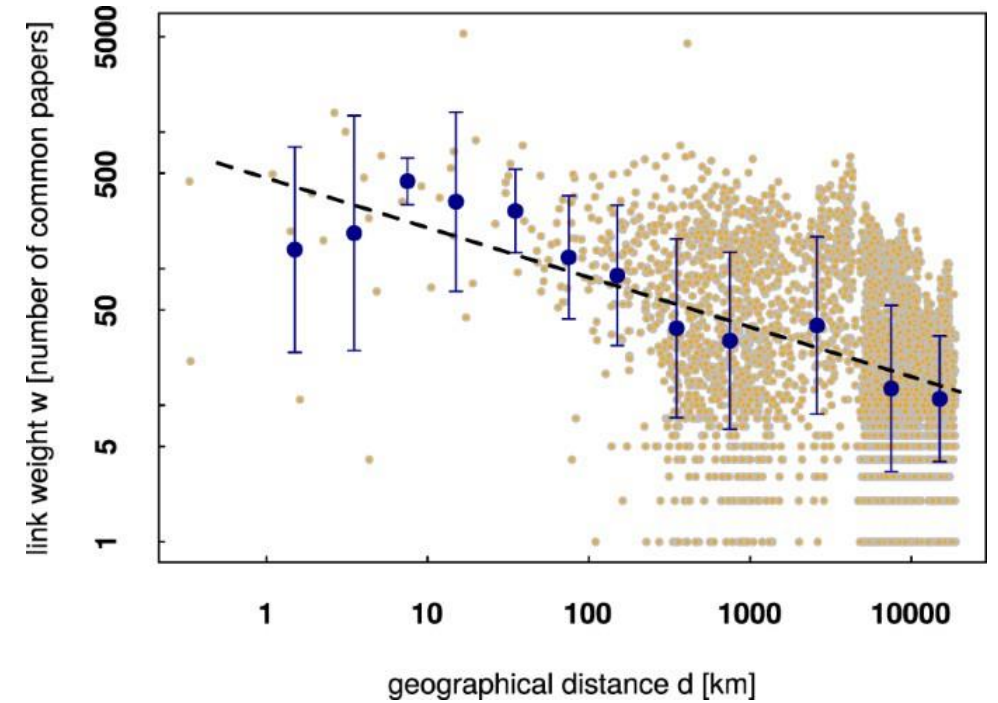
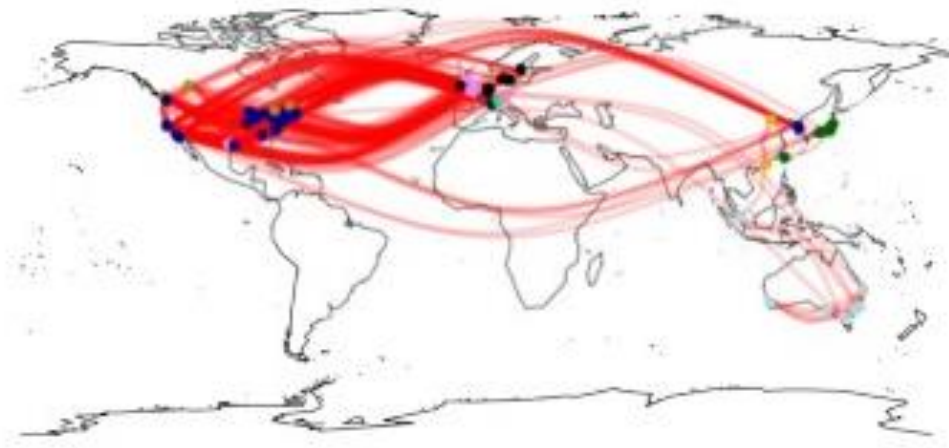
$L(d)$ – observed connections

$P(d) = L(d) / N(d)$



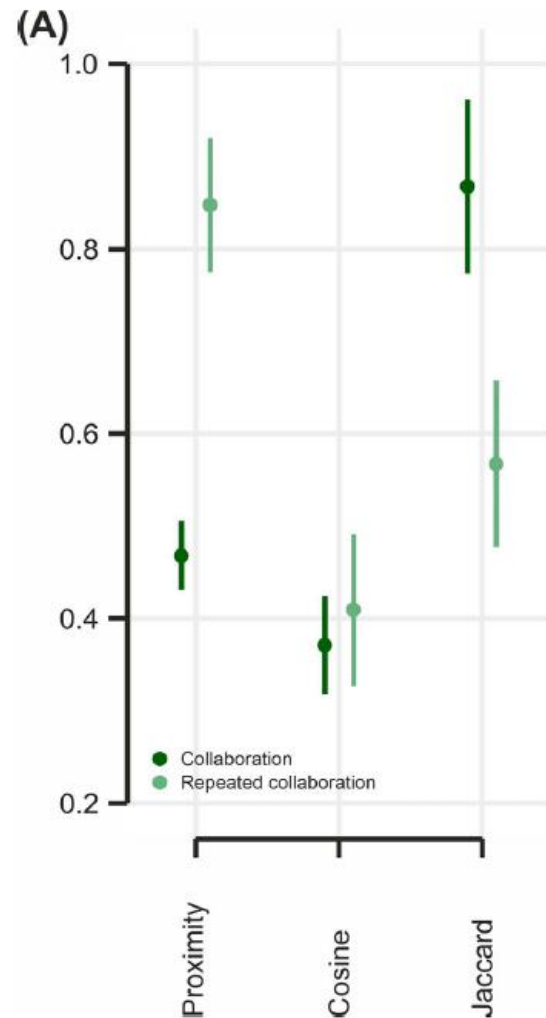
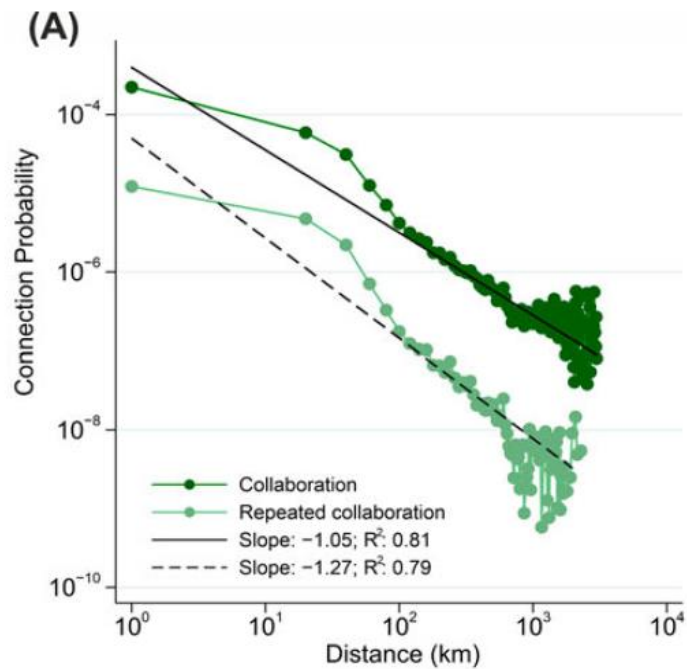
Lengyel B, Varga A, Ságvári B, Jakobi Á, Kertész J (2015) Geographies of an Online Social Network. PLoS ONE 10(9): e0137248. doi:10.1371/journal.pone.0137248
<http://127.0.0.1:8081/plosone/article?id=info:doi/10.1371/journal.pone.0137248>

The gravity law



Sienkiewicz, J., Soja, K., Hołyst, J.A. *et al.* Categorical and Geographical Separation in Science. *Sci Rep* **8**, 8253 (2018).
<https://doi.org/10.1038/s41598-018-26511-4>

The gravity regression



$$\Pr(Y_{ij,t}) = \gamma_0 + \gamma_1 \theta_{ij} + \gamma_2 (Z_{i,t-1}, Z_{j,t-1}) + \varepsilon_{ij,t}$$

$$\log(Y_{ij,t} = y_{ij,t}) = \beta_0 + \beta_1 P_{ij} + \beta_2 C_{ij,t-1} + \beta_3 J_{ij,t-1} + \varepsilon_{ij,t}$$

Tóth et al. 2021 European Planning Studies

Urban scaling

$$Y(t) = Y_0 N(t)^\beta.$$

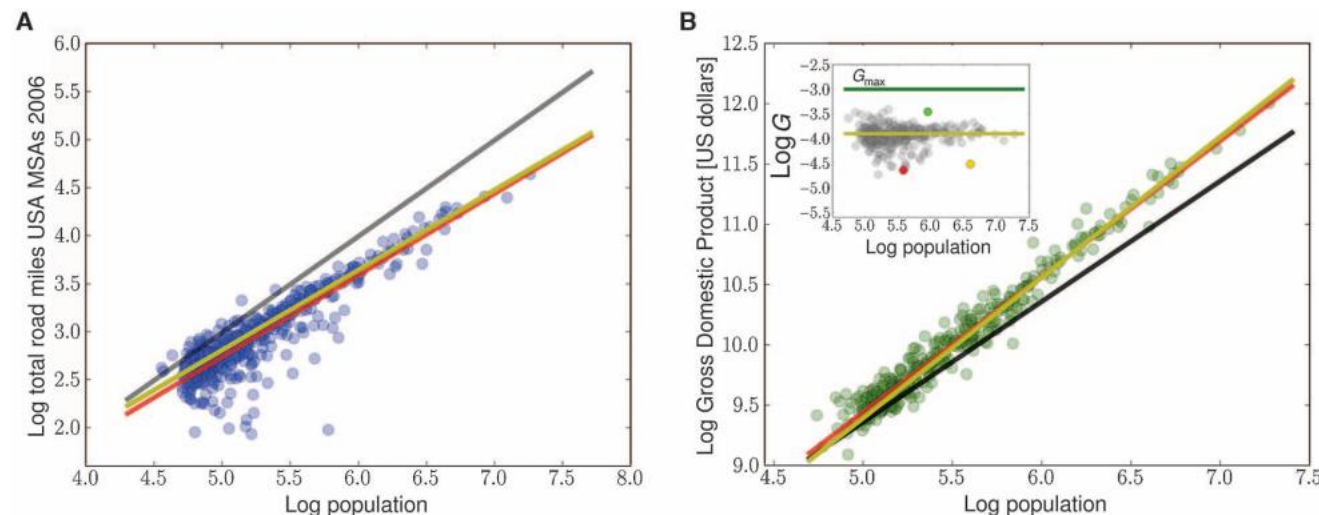


Fig. 1. Scaling of urban infrastructure and socioeconomic output. (A) Total lane miles (volume) of roads in U.S. metropolitan areas (MSAs) in 2006 (blue dots). Data for 415 urban areas were obtained from the Office of Highway Policy Information from the Federal Highway Administration (14). Lines show the best fit to a scaling relation $Y(N) = Y_0 N^\beta$ (red), with $\beta = 0.849 \pm 0.038$ [95% confidence interval (CI), $R^2 = 0.65$]; the theoretical prediction, $\beta = 5/6$ (yellow); and linear scaling $\beta = 1$ (black). **(B)** Gross metropolitan product of MSAs in 2006 (green dots). Data obtained for 363 MSAs from U.S. Bureau of Economic Analysis (14). Lines describe best fit (red) to data, $\beta = 1.126 \pm 0.023$ (95% CI, $R^2 = 0.96$); the theoretical prediction, $\beta = 7/6$ (yellow); and proportional scaling, $\beta = 1$ (black). The two best-fit parameters in each scaling

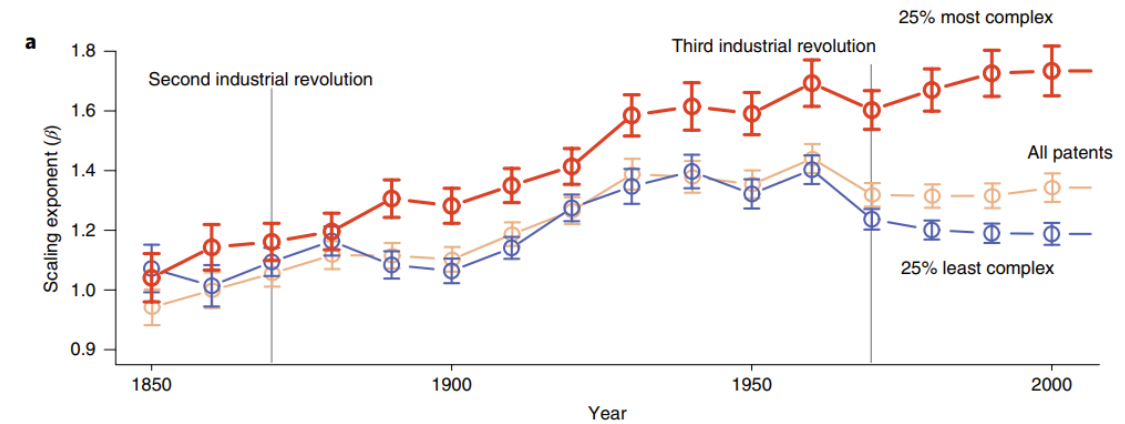
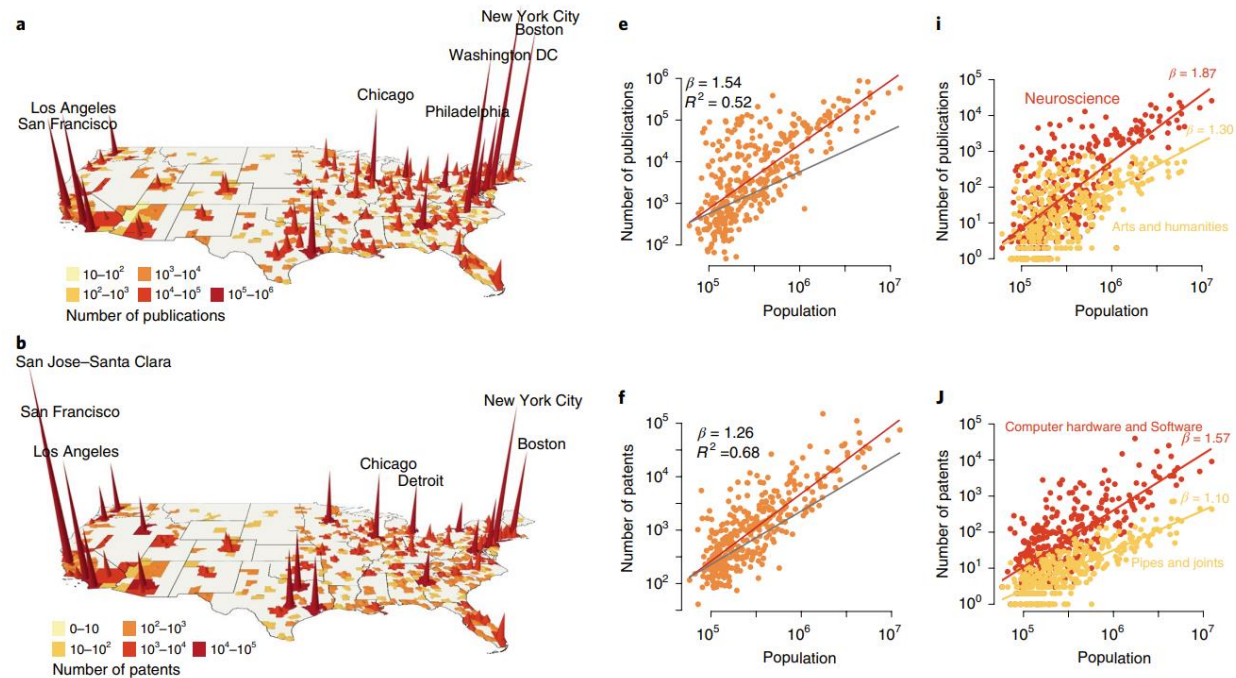
relation were estimated by means of ordinary least-squares minimization to the linear relation between logarithmically transformed variables (14). The inset shows the estimate of G for 313 U.S. MSAs and the conservation law $\frac{d \ln G}{d \ln N} = 0$ ($R^2 = 0.003$). G is measured as the product of gross domestic product and road volume, both per capita. As predicted by the theory, observed values of G for different cities cluster around its most likely value (mode, yellow line), which gives an estimate of the optimum G^* , and are bounded by the maximum $G_{\max} \approx 8G^*$ (green line); see also Fig. 2B. Several metropolitan areas, such as Bridgeport, Connecticut (green circle); Riverside, California (yellow circle); or Brownsville, Texas (red circle), are outliers, suggesting that they are suboptimal in terms of their transportation efficiency or amount of social mixing.

Table 1. Scaling exponents for urban indicators vs. city size

Y	β	95% CI	Adj- R^2	Observations
New patents	1.27	[1.25, 1.29]	0.72	331
Inventors	1.25	[1.22, 1.27]	0.76	331
Private R&D employment	1.34	[1.29, 1.39]	0.92	266
"Supercreative" employment	1.15	[1.11, 1.18]	0.89	287
R&D establishments	1.19	[1.14, 1.22]	0.77	287
R&D employment	1.26	[1.18, 1.43]	0.93	295
Total wages	1.12	[1.09, 1.13]	0.96	361
Total bank deposits	1.08	[1.03, 1.11]	0.91	267
GDP	1.15	[1.06, 1.23]	0.96	295
GDP	1.26	[1.09, 1.46]	0.64	196
GDP	1.13	[1.03, 1.23]	0.94	37
Total electrical consumption	1.07	[1.03, 1.11]	0.88	392
New AIDS cases	1.23	[1.18, 1.29]	0.76	93
Serious crimes	1.16	[1.11, 1.18]	0.89	287
Total housing	1.00	[0.99, 1.01]	0.99	316
Total employment	1.01	[0.99, 1.02]	0.98	331
Household electrical consumption	1.00	[0.94, 1.06]	0.88	377
Household electrical consumption	1.05	[0.89, 1.22]	0.91	295
Household water consumption	1.01	[0.89, 1.11]	0.96	295
Gasoline stations	0.77	[0.74, 0.81]	0.93	318
Gasoline sales	0.79	[0.73, 0.80]	0.94	318
Length of electrical cables	0.87	[0.82, 0.92]	0.75	380
Road surface	0.83	[0.74, 0.92]	0.87	29

Complex economic activities concentrate in large cities

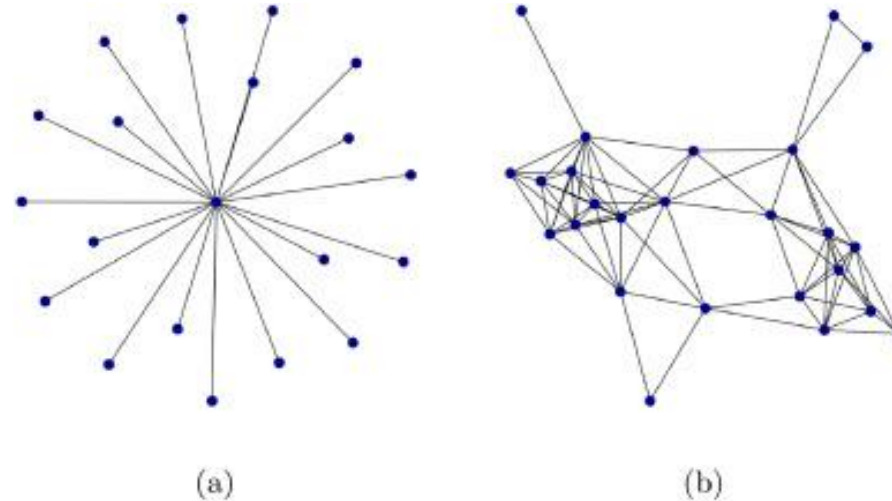
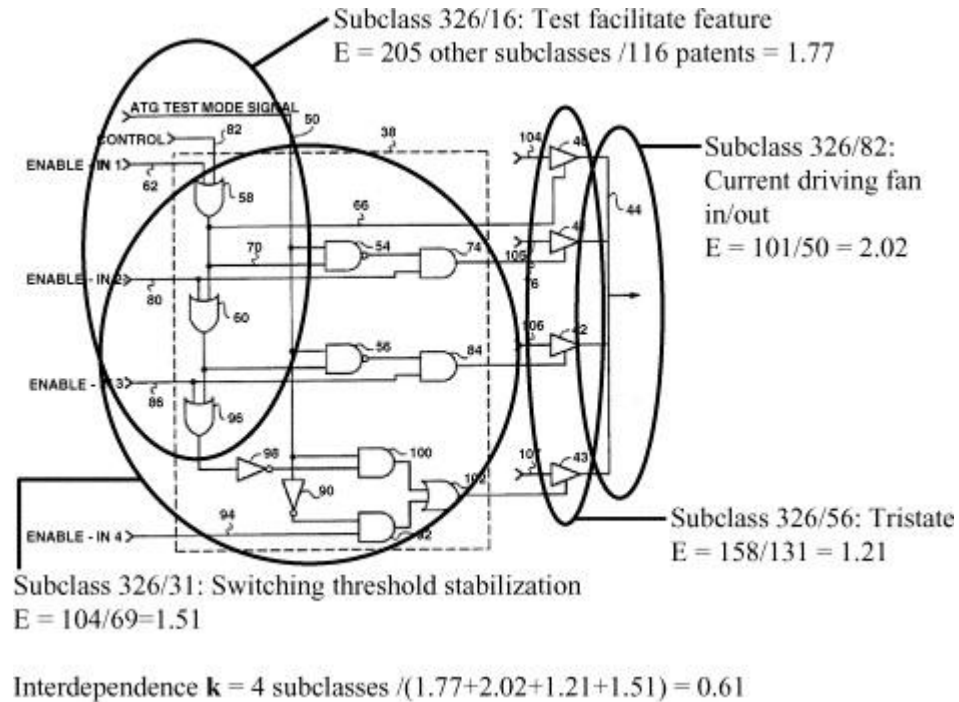
Pierre-Alexandre Balland ^{1,2*}, Cristian Jara-Figueroa ², Sergio G. Petralia ³, Mathieu P. A. Steijn ^{1,4}, David L. Rigby ⁵ and César A. Hidalgo ^{2,6,7,8}



Definition of complexity of knowledge

Activity	Proxy for complexity	Intuition
Technologies / patents	Average year in which patent subclasses were introduced	Newer technologies tend to combine a larger accumulated stock of knowledge
Scientific fields	Average number of authors per publication	More complex knowledge production requires larger teams / more specialized knowledge
Industries	Average years of education of workers	More knowledge-intensive industries require more accumulated human capital
Occupations	Average years of education of workers	More complex occupations require more accumulated knowledge

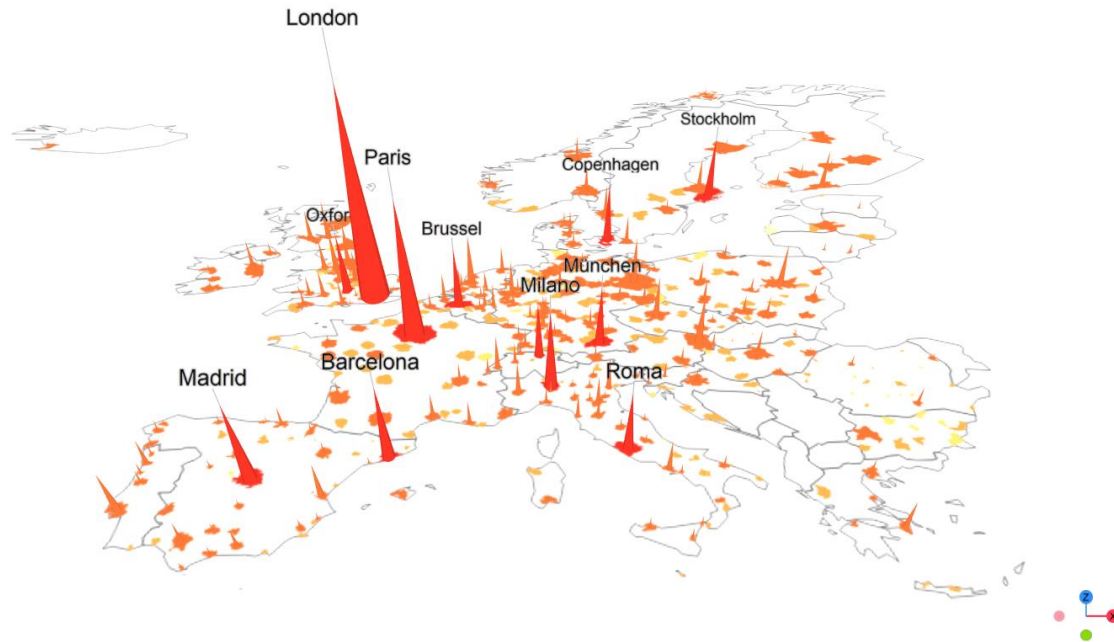
Combination of knowledge as measurement of knowledge complexity



Sorenson et al. 2006, Research Policy; Mewes and Broekel 2002, Research Policy

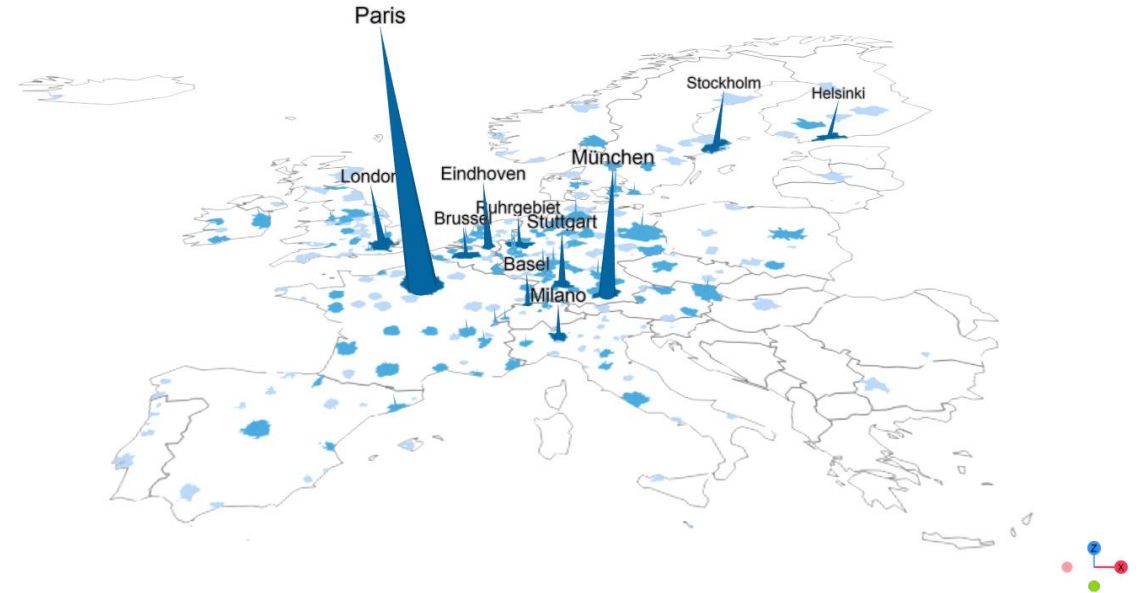
Geography of Knowledge Production in Europe, 2020

A. Scientific papers



Dominated by Paris, London, Madrid, Rome, Vienna — follows population.

B. Patents



Paris, Munich, Eindhoven — concentrated in the 'blue banana' corridor.

Data

Sample

408 European Functional Urban Areas (FUAs) hosting ≥ 1 university · 1980–2020

Scientific papers

Scopus API — all EU universities indexed in 2020 · 3-year averages

Patents

OECD REGPAT 2024 · geocoded at 98% (tidygeocoder + Eurostat ZIP + city name)

University age

EHESO database · 1,266 universities · 37 countries · year of establishment

EU funding

CORDIS · FP7 + Horizon2020 grant contributions · aggregated to FUA level

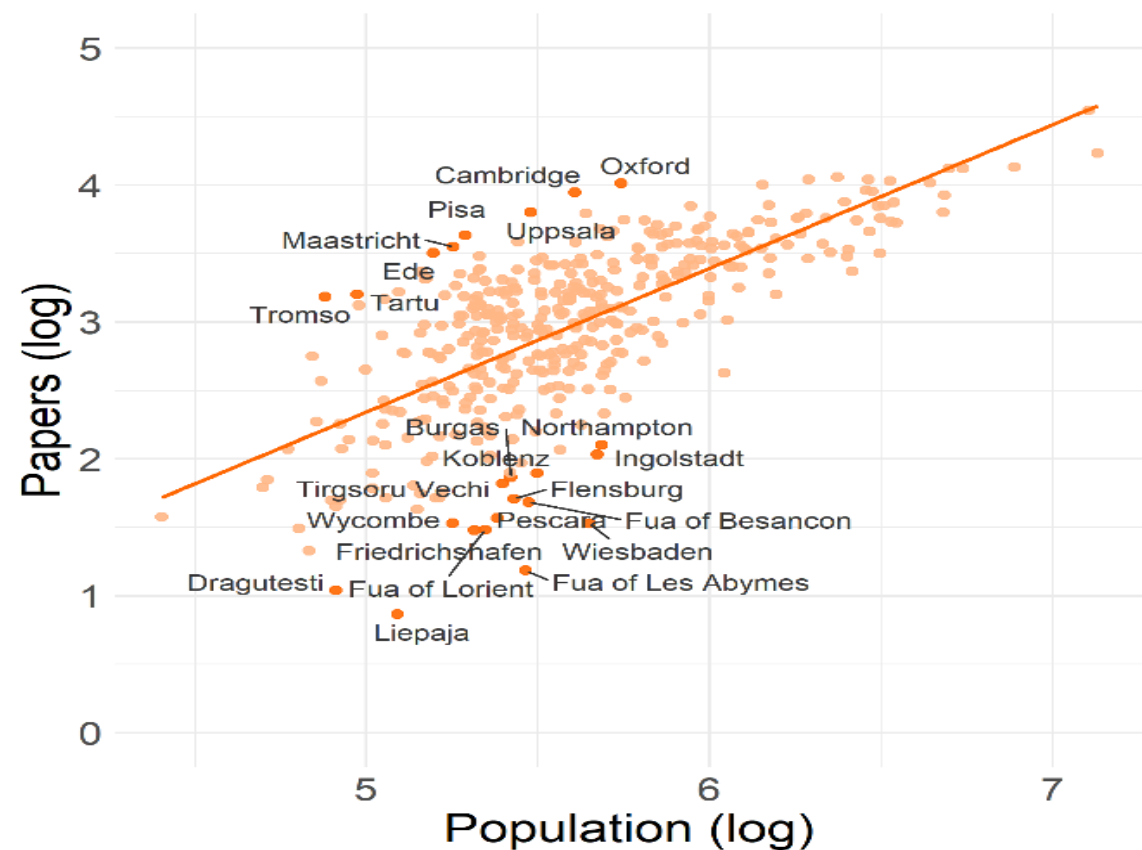
Population

CIESIN 2021 gridded density (1970–2020) + HYDE 3.1 historical (1000 AD)

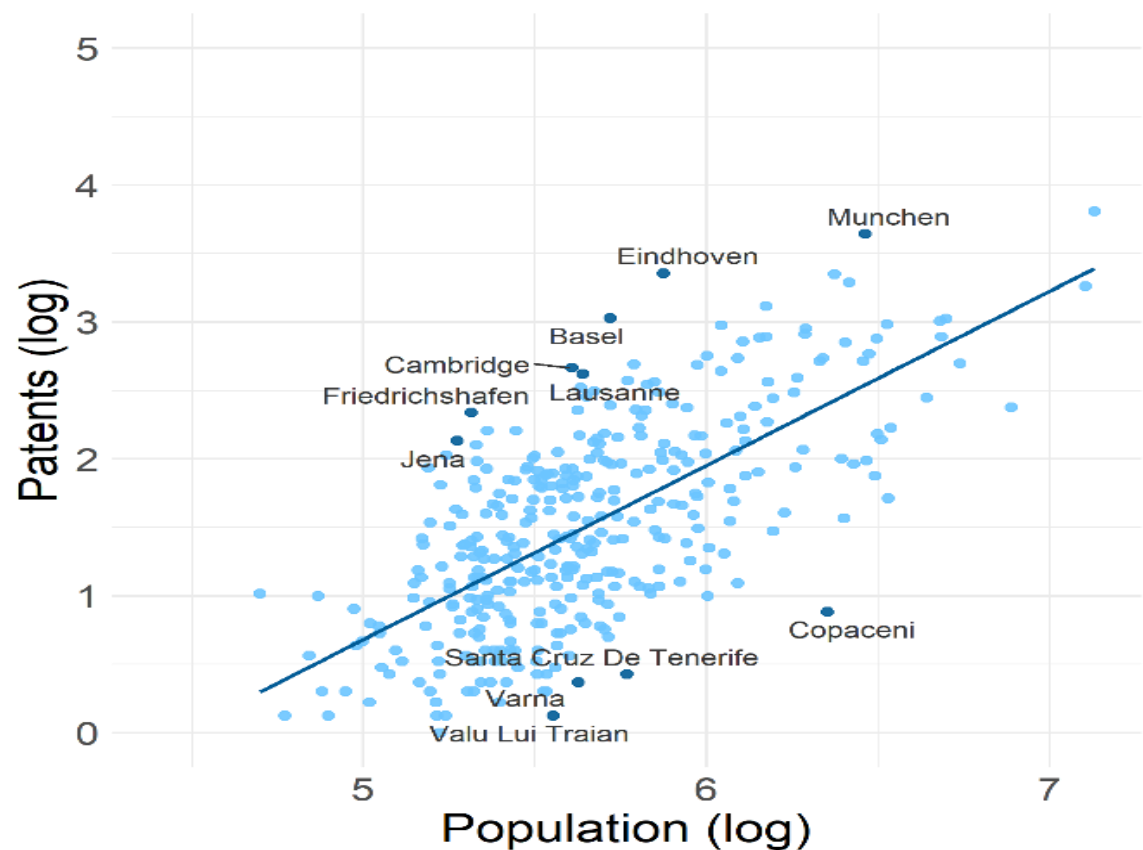
University age classification follows Ridder-Symoens (1992) and Rüegg (1996, 2004, 2011): Medieval ≤ 1500 , Early Modern 1501–1800, Industrial Era 1801–1945, Post-War 1946–1989, Contemporary 1990–present.

Urban Scaling of Knowledge Production, 2020

C. Scientific papers ($\beta = 1.049$, $R^2 = 0.49$)



B. Patents ($\beta = 1.317$, $R^2 = 0.46$)

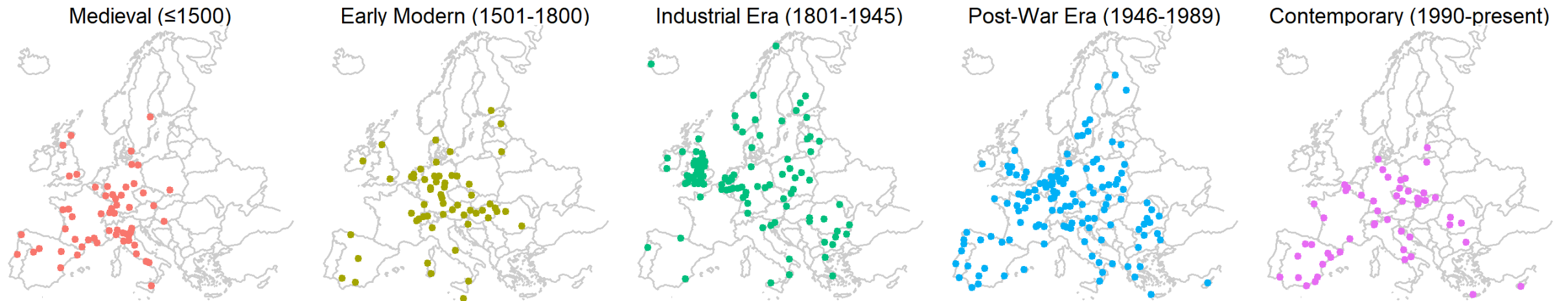


Papers scale nearly linearly. Patents scale superlinearly. Small/medium cities with old universities (Oxford, Cambridge, Tartu) punch above weight for papers; Eastern European cities underperform in patents.

**University founding era and research output in 2020,
OLS estimates.** Standard errors in parentheses. ***p<0.01,
**p<0.05, *p<0.10.

	Papers – university	Papers – city	Patents – city
	(1)	(2)	(3)
Reference: Medieval (≤1500)			
Early Modern (1501–1800)	-1,102.267*** (216.200)	-218.116 (528.984)	-58.257 (98.126)
Industrial Era (1801–1945)	-1,725.359*** (182.140)	-2,305.718*** (497.778)	-101.237 (93.673)
Post-War Era (1946–1989)	-2,061.006*** (176.142)	-2,973.109*** (444.067)	-123.761 (83.598)
Contemporary (1990–present)	-2,291.725*** (183.908)	-3,288.175*** (542.359)	-196.060* (107.993)
Constant	2,154.937*** (277.712)	3,419.540*** (1,161.414)	258.576 (213.884)
Observations	875	408	366
R ²	0.320	0.247	0.074
Residual Std. Error	1,204.642	2,671.781	491.592
F Statistic	11.643***	3.601***	0.803

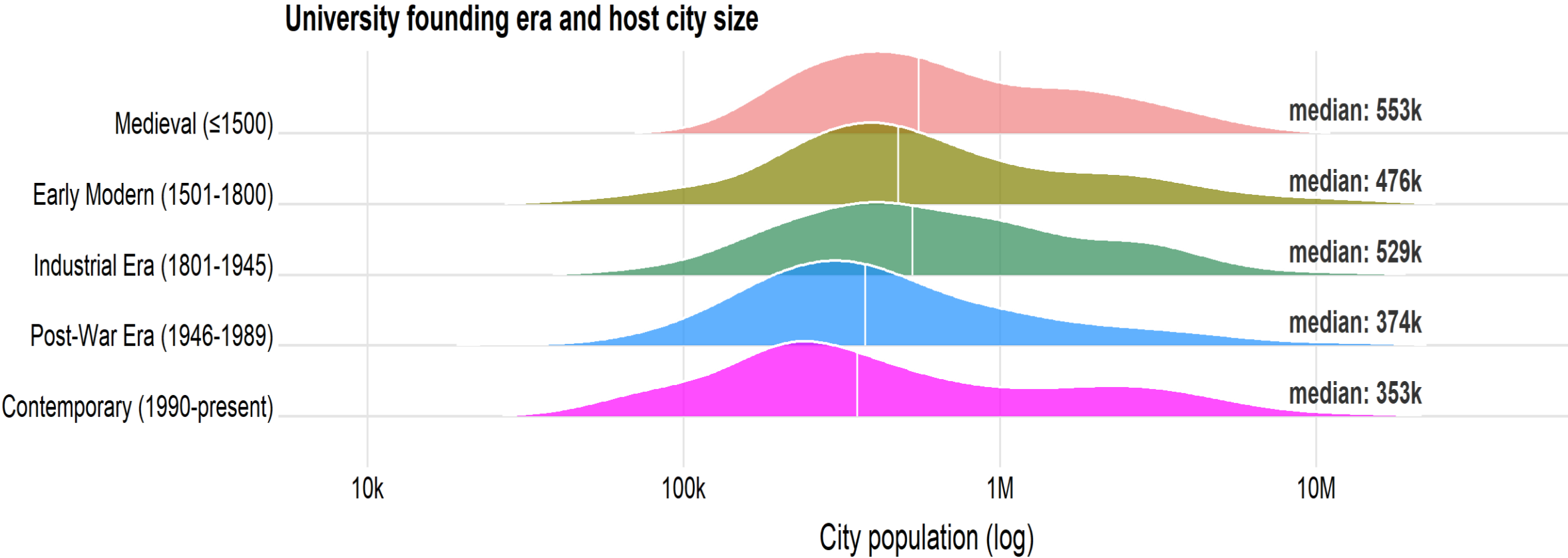
Geography of University Founding Eras



Medieval universities (≤ 1500) cluster in Western and Southern Europe. Industrial Era expansion moves to Britain and Germany. Post-War and Contemporary universities spread to Eastern Europe and the periphery.

Source: EHESO database (European Commission, 2025). 408 FUAs classified by founding era of the oldest university in the city.

University Founding Era and Host City Size

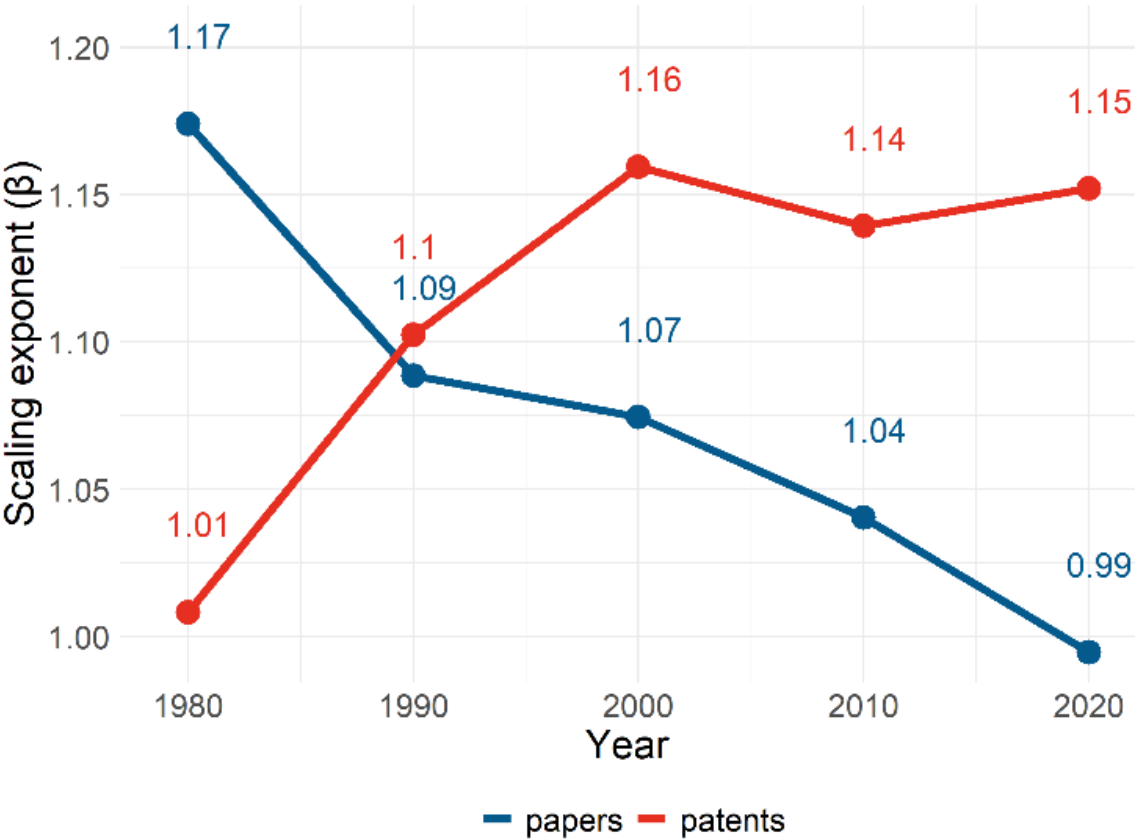


Post-War and Contemporary universities are systematically located in smaller cities (median 374k and 353k) compared to Medieval and Industrial Era cities (553k and 529k).

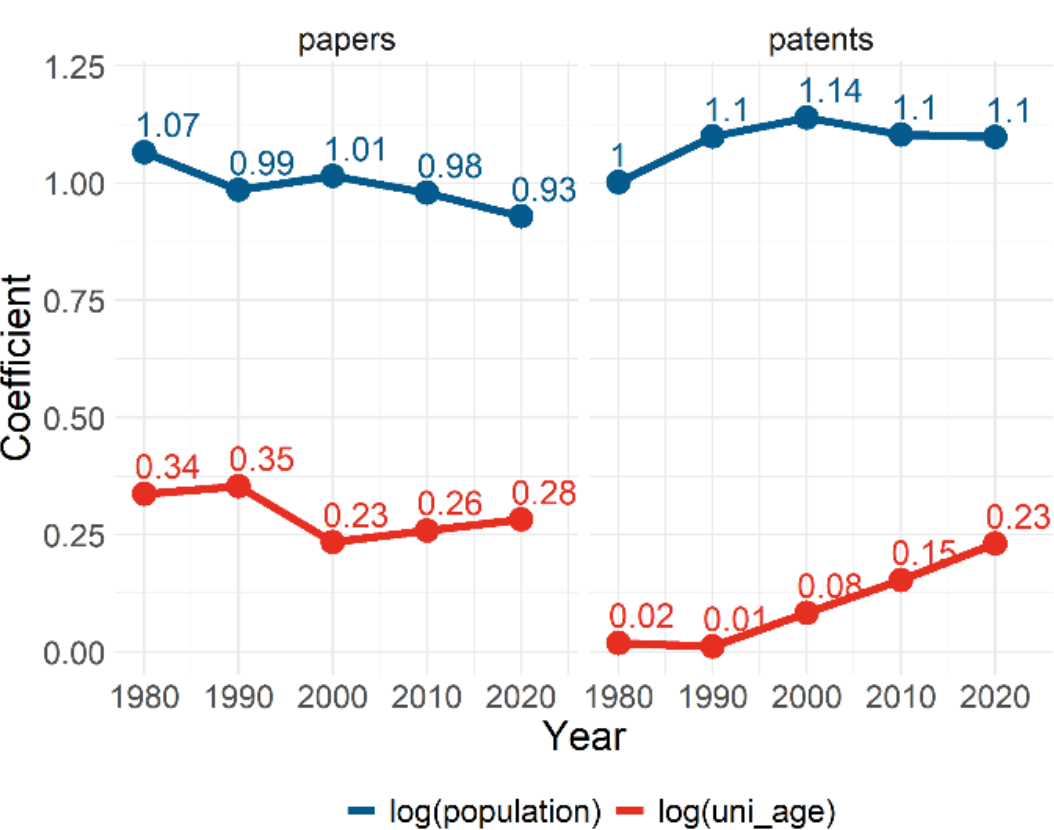
Pairwise tests confirm Post-War and Contemporary city sizes are significantly smaller than earlier eras ($p < 0.05$).

Urban Scaling Dynamics, 1980–2020 (113 cities with papers and patents in all years)

A. Scaling exponent β over time



B. Coefficients controlling for university age



Papers: β falls from 1.17 (1980) to 0.99 (2020) — deconcentration. Patents: β rises from 1.01 to 1.15 — growing concentration. University age effect on patents is rising (0.02→0.23) — history matters more over time.

Fixed-Effects Panel regressions (City FE + Period FE)

$$Pat_{i,t} = \alpha_c + \beta_1 Pop_{i,t-1} + \beta_2 Publ_{i,t-1} + \beta_3 Pop_{i,t-1} \times Publ_{i,t-1} + u_t + \varepsilon_{i,t},$$

$$\frac{\delta Pat}{\delta Publ} = \beta_2 + \beta_3 Pop$$

$$Pat_{i,t} = \alpha_c + \beta_1 Pop_{i,t-1} + \beta_2 Publ_{i,t-1} + \beta_3 Age_i \times Publ_{i,t-1} + u_t + \varepsilon_{i,t}$$

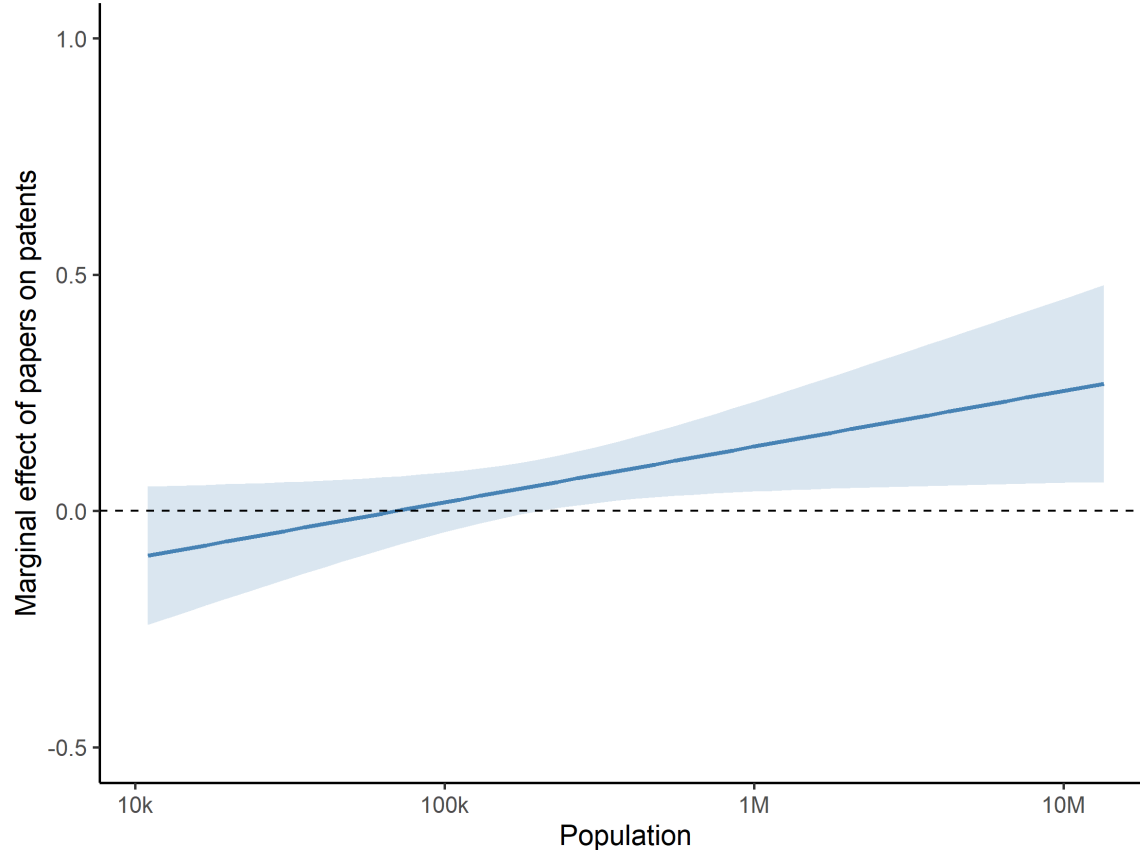
Science, University Age and Patenting — Fixed-Effects Panel (City FE + Period FE)

	Pat.(1)	Pat.(2)	Pat.(3)	Pat.(4)	Pap.(5)
Population (log, lag)	0.767+	0.464	0.722+	0.697+	0.345+
	(0.404)	(0.450)	(0.395)	(0.388)	(0.179)
Papers (log, lag)	0.068*	−0.524	0.183**	0.176**	0.609***
	(0.034)	(0.326)	(0.057)	(0.056)	(0.160)
Pop. × Papers		0.111+			−0.045
		(0.060)			(0.029)
Papers × Early Modern			−0.017	−0.018	
Papers × Industrial Era			−0.057	−0.056	
Papers × Post-War			−0.105+	−0.106+	
Papers × Contemporary			−0.135*	−0.123*	
University count (log, lag)				0.402***	
N	1,297	1,297	1,297	1,297	1,659
R ² (within)	0.015	0.024	0.037	0.051	0.332

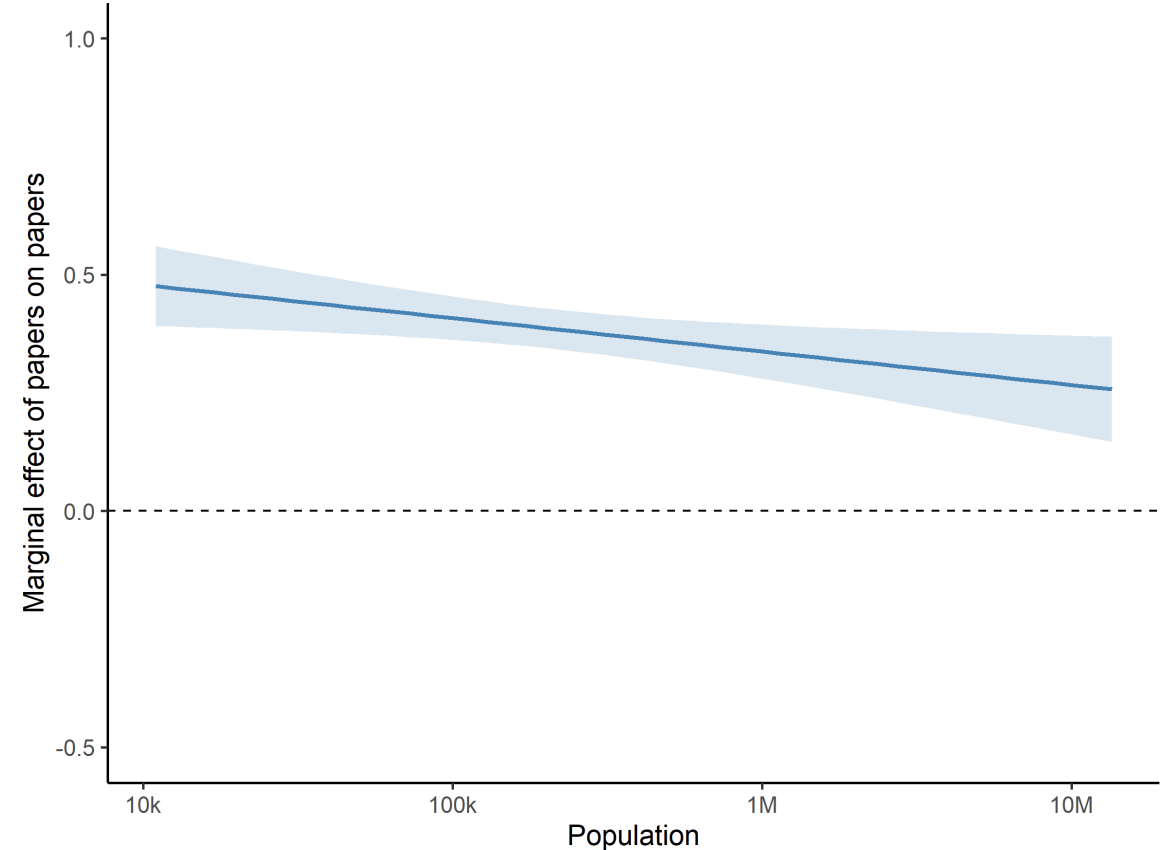
Clustered SE by URAU_CODE. + p<0.10 * p<0.05 ** p<0.01 *** p<0.001. Models 1–4: N=1,297 (cities with patents); Model 5: N=1,659 (full sample including cities with papers only).

Marginal Effect of Scientific Papers on Patents and Papers by City Population

A. Papers → Patents (significant only in large cities)



B. Papers → Papers (significant at all city sizes)



The science→patent link is insignificant for cities below ~300k and grows with city size (A). Scientific knowledge compounds across all city sizes, slightly stronger in small cities — consistent with deconcentration (B).

Patent Growth regression — Cross-Section OLS (Country FE)

$$\begin{aligned} &\delta Pat_{i,2000 \rightarrow 2020} \\ &= \alpha_c + \beta_1 Pop_{i,2000} + \beta_2 Publ_{i,2000} + \beta_3 Patents_{i,t2000} + \beta_4 EU Funds_i \\ &+ \beta_5 Age_i + \beta_6 EU Funds_i \times \beta_3 Age_i + c_i + \varepsilon_i \end{aligned}$$

Patent Growth 2000–2020 and EU Funding — Cross-Section OLS (Country FE)

	(1)	(2)	(3)	(4)
Population 2000 (log)	0.275***	0.275***	0.267***	0.274***
	(0.062)	(0.063)	(0.064)	(0.063)
Papers 2000 (log)	0.083***	0.083***	0.066**	0.077**
	(0.029)	(0.030)	(0.032)	(0.031)
EU Funds (log)		0.000	−0.013	−0.006
		(0.008)	(0.013)	(0.009)
EU Funds × Early Modern			0.010	
EU Funds × Industrial Era			0.013	
EU Funds × Post-War Era			0.023*	
EU Funds × Contemporary			0.032	
EU Funds × Post-WWII				0.020**
				(0.010)
N	302	302	302	302
R ²	0.462	0.462	0.481	0.472

Dependent variable: $\Delta \log_{10}(\text{patents}_{2020} - \text{patents}_{2000})$. Country FE in all models. Baseline patents included but not shown. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Conclusions

1. Diverging spatial trajectories

Patent production is increasingly concentrated in large cities (β : 1.01→1.15); scientific publication is deconcentrating (β : 1.17→0.99). A widening spatial mismatch between where knowledge is produced and where it is commercialised.

2. History shapes knowledge convertibility

University age independently predicts both the volume of scientific output and the strength of the science-to-patent link, beyond city size. Cities with medieval universities outperform — through accumulated institutional depth, not agglomeration alone.

3. Young universities face a double constraint

Smaller cities with post-war universities have both weaker agglomeration economies and a shorter history of industry-academia linkages. Scientific capacity can be built quickly; absorptive capacity cannot.

4. EU funding helps — but risks a territorial innovation trap

EU R&D funding boosts patenting specifically in post-1945 university cities (Post-War and Contemporary groups). However, funding science without valorisation infrastructure risks expanding research inputs while conversion capacity stagnates (Diemer et al., 2022; Boschma et al., 2025).